

NEPHROLOGY TEACHING SERIES

Acute Kidney Injury

A structured approach to the AKI consult

Creatinine is not the consult — trajectory, urine output, and reversible factors are.

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MS3/MS4 rotation teaching

Why we get consulted

The consult is a question about trajectory and reversibility, not a creatinine number.

TRIGGERS FOR NEPHROLOGY CONSULT

- Rising Cr without clear cause
- Stage 2–3 by KDIGO
- Active sediment (RBC / muddy brown casts)
- Approaching KRT — acidosis, electrolytes, ingestion, overload, uremia (AEIOU)
- Cirrhosis, HF, or sepsis complicating AKI

TEACHING PEARL

Creatinine is a symptom. Spend your first 5 minutes on trajectory, hemodynamics, and reversible contributors — not the number itself.

KDIGO staging: creatinine or urine output

Stage by **WHICHEVER** is worse — the UOP criterion is often the first to trip.

Stage	Serum Creatinine	Urine Output
1	1.5–1.9× baseline OR $\uparrow \geq 0.3$ mg/dL within 48 h	< 0.5 mL/kg/h for 6–12 h
2	2.0–2.9× baseline	< 0.5 mL/kg/h for ≥ 12 h
3	$\geq 3\times$ baseline OR Cr ≥ 4.0 mg/dL OR initiation of KRT	< 0.3 mL/kg/h for ≥ 24 h OR anuria ≥ 12 h

Why UOP matters: oliguria often precedes the creatinine rise by 24–48 h. Review bladder scan / Foley output hour-by-hour, not just the 24-hr total.

Anatomic differential: the three buckets

Every AKI gets triaged into these three buckets on the first pass.

PRE-RENAL (perfusion)



Hypovolemia: GI, diuretic, bleed
Sepsis, HF, cirrhosis
ACEi/ARB + NSAID + dehydration
BUN/Cr > 20, FeNa < 1%, FeUrea < 35%
Reverses with perfusion

INTRINSIC (parenchymal)



ATN: ischemia, sepsis, contrast, pigment
AIN: PPI, abx, NSAID
GN: RBC casts, proteinuria
Vascular: TMA, emboli, infarct
Does NOT respond to volume

POST-RENAL (obstruction)



BPH, pelvic mass, neurogenic bladder
Bilateral ureteral: stones, retroperitoneal
Crystal: acyclovir, MTX, tumor lysis
Ultrasound is the screening test
Post-relief: watch K, Mg, PO4

Workup: the first-pass labs

If you skip the microscopy, you'll miss the diagnosis that changes management.

MINIMUM DATASET

BMP + Cr trend (all prior values)

UA + microscopy

UPCR or UACR

Urine Na, Cr, urea → FeNa, FeUrea

Renal US (obstruction)

CBC, LFTs, CK, lactate

Med review: NSAIDs, RAAS, contrast

Fractional excretion interpretation

Test	Pre-renal	ATN
FeNa	< 1%	> 2%
FeUrea	< 35%	> 50%
BUN/Cr	> 20	10–15
Urine osm	> 500	≈ 300 (iso-)
UA	bland	muddy brown casts

FeNa is unreliable on diuretics. Use FeUrea instead (urea reabsorption is not blocked by loop/thiazide).

What the lab UA tells you

No spun microscopy needed — the dipstick + automated report answers most questions.

READ THE DIPSTICK

Color — tea/coke = myoglobin or hemoglobin

Specific gravity — > 1.020 concentrated (prerenal); ≈ 1.010 fixed (ATN)

Protein — trace to 3+ (quantify with UPCr / UACr)

Blood — hematuria; confirm with RBC count

Leukocyte esterase / nitrite — UTI vs AIN

Glucose / ketones — DKA, SGLT2i euglycemic DKA

DISCREPANCIES THAT MEAN SOMETHING

Blood+ on dipstick, no RBCs → myoglobin (rhabdo) or hemoglobin

Protein+ with bland UA → glomerular — get UPCr

LE+ but nitrite neg → AIN or fastidious organism

Nitrite+ → gram-negative UTI (usually *E. coli*)

Negative dipstick in AKI does NOT rule out GN

Always pair the UA with: spot UPCr / UACr (quantify), BMP (Cr trend, HCO₃, K), urine Na/Cr/urea (FeNa / FeUrea).

When to start KRT: the AEIOU mnemonic

Indications are clinical, not a single lab threshold.



Acidosis

pH < 7.1–7.15 refractory to bicarbonate/ventilation.



Electrolytes

K⁺ > 6.5 or with ECG changes refractory to medical therapy.



Ingestion

Dialyzable toxin: methanol, ethylene glycol, salicylates, lithium, metformin (MALA).



Overload

Refractory pulmonary edema; need for aggressive nutrition or blood products.



Uremia

Pericarditis, encephalopathy, bleeding, intractable nausea/pruritus.

STARRT-AKI & AKIKI: without a clear AEIOU indication, watchful waiting is safe — ~40% of delayed-strategy patients recovered without ever needing KRT.

Management principles

Remove the insult. Restore perfusion. Avoid the second hit.

STOP NEPHROTOXINS

Hold ACEi/ARB, NSAIDs, aminoglycosides. Renal-dose the rest.
Defer contrast if possible.

RESTORE PERFUSION

Volume only if depleted. Balanced crystalloids > NS (SMART).
Sepsis: MAP $\geq$ 65.

TREAT THE CAUSE

Abx for sepsis. Relieve obstruction. Stop AIN trigger.
Immunosuppression for GN/vasculitis.

SUPPORT, DON'T OVERCORRECT

Watch K, HCO₃, PO₄. Diurese only if overloaded — not to chase Cr.

Landmark trials that shape practice

What changed, and what you're expected to know on rounds.

Trial	Year	Takeaway
ATN Trial	2008	Intensive CRRT (35 mL/kg/h) = standard (20–25). Ended 'more is better.'
AKIKI	2016	Delayed KRT safe in stage 3 AKI; 49% never needed dialysis.
STARTRT-AKI	2020	Accelerated KRT did not reduce 90-d mortality; 38% of standard arm avoided KRT entirely.
SMART	2018	Balanced crystalloids reduced major adverse kidney events at 30 days (MAKE30) vs NS in ICU (NNT 91; NNT 29 in sepsis).
BICAR-ICU	2018	IV HCO ₃ ⁻ for pH ≤ 7.20 reduced mortality in the AKI subgroup (NNT ~6).
PRESERVE	2018	NAC does NOT prevent contrast AKI; isotonic saline = bicarbonate. Debunked two dogmas.

Common presentation mistakes

Avoid these and you'll sound like a senior.

AVOID

- Leading with Cr value, not trajectory
- Reporting 24-hr UOP, not hourly
- Calling every AKI 'ATN' without UA
- Missing RAAS + NSAID + contrast combo
- Recommending KRT by labs alone
- Skipping US in anuria

DO INSTEAD

- Open: stage, trajectory, UOP, top 2 ddx
- Pull ≥ 3 prior Cr values
- Always state urine microscopy result
- Name one reversible factor you're removing
- Frame KRT by AEIOU, not BUN/Cr
- Anuria $\rightarrow$ US reflexively

Clinical case

Think it through — answer is on the next slide.

VIGNETTE

A 72-year-old man with HTN, CHF (EF 30%), and baseline Cr 1.4 is admitted for pneumonia. Started on IV piperacillin-tazobactam and NS boluses. On hospital day 3, Cr is 2.8, UOP 0.3 mL/kg/h over 12 h. BP 108/65, on room air. UA: 2+ protein, granular/muddy-brown casts. Meds held: lisinopril, furosemide. Ultrasound: no hydronephrosis.

QUESTION: What stage of AKI is this, and what's your top differential?

Clinical case — answer

Key teaching points from the case.

ANSWER

KDIGO stage 2 AKI (Cr 2× baseline); most likely ATN from sepsis ± piperacillin-tazobactam synergistic nephrotoxicity.

TEACHING

Muddy-brown casts and a non-response to volume point to intrinsic ATN rather than pre-renal. Stage is driven by both Cr (2× baseline = stage 2) and UOP (< 0.5 for ≥ 12 h = stage 2). Plan: continue holding RAAS + nephrotoxins, optimize MAP ≥ 65, watch K⁺/HCO₃ closely, daily UA, consider biopsy only if sediment unusual or failure to recover by day 7.

REFERENCES

Acute Kidney Injury

Guidelines

KDIGO 2012 Clinical Practice Guideline for AKI
Surviving Sepsis Campaign 2021 sepsis guidelines
ACR Manual on Contrast Media (contrast-associated AKI)
ASN educational resources

Landmark Trials

ATN Trial — NEJM 2008
AKIKI — NEJM 2016
STARRT-AKI — NEJM 2020
SMART — NEJM 2018
SALT-ED — NEJM 2018
BICAR-ICU — Lancet 2018
PRESERVE — NEJM 2018
AMACING — Lancet 2017

UpToDate (2026 access):

Definition and staging criteria of AKI in adults • Evaluation of AKI among hospitalized adult patients • Overview of management of AKI in adults • KRT in AKI: indications, timing, and dose